

A Level Computer Science (9618)

Topical Past Paper Worksheet

Topic 3: Hardware

Subtopic 3.1: Computers and their Components

Concept 3.1.2: Embedded Systems

Total Questions: 8

Mark Schemes begin on Page 5

Question 1 | Source: 9618_w24_qp_12 (Q2.a)

2 Embedded systems are used in many electronic devices.

(a) Describe the drawbacks of embedded systems.

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..... [3]

Question 2 | Source: 9618_s24_qp_11 (Q2.a)

- 2 A video doorbell is attached to the front door of a house. The doorbell uses a motion sensor to detect when a visitor walks in front of the door. When the motion sensor is activated:
- The digital camera in the doorbell starts recording a video.
 - A message is transmitted to a smartphone so that the person who lives in the house can watch the video.

The doorbell also has a button that can be pressed. When the button is pressed, a message is transmitted to a smartphone to play the doorbell sound.

The videos are stored on the doorbell's internal secondary storage device and overwritten when the secondary storage device is full.



(a) The video doorbell can be considered an example of an embedded system.

Identify **two** characteristics of the doorbell that suggest it is an embedded system.

1

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2

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[2]

Question 3 | Source: 9618_w23_qp_11 (Q9.b)

- (b) Give **one** example of an embedded system **and** explain why it is an example of an embedded system.

Example

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Explanation

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[3]

Question 4 | Source: 9618_w23_qp_12 (Q1.c.i)

- 1 A factory makes chocolate bars.

The factory uses a conveyor belt that moves the products from one stage of production to the next stage.

- (c) The factory has many different machines with embedded systems.

- (i) Identify **two** features of embedded systems.

1

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2

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[2]

Question 5 | Source: 9618_w23_qp_12 (Q1.c.ii)

- 1 A factory makes chocolate bars.

The factory uses a conveyor belt that moves the products from one stage of production to the next stage.

- (c) The factory has many different machines with embedded systems.

(ii) Identify **one** drawback of embedded systems.

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..... [1]

Question 6 | Source: 9618_w22_qp_11 (Q9.a)

9 Many modern televisions are examples of embedded systems.

(a) Explain why these televisions are embedded systems.

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..... [2]

Question 7 | Source: 9618_w22_qp_13 (Q10.b.ii)

(ii) The car alarm is an example of an embedded system.

Describe the characteristics of an embedded system.

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..... [3]

Question 8 | Source: 9618_s21_qp_11 (Q5.a)

5 Kiara has a washing machine and a refrigerator.

(a) She has an embedded system in her washing machine.

Describe what is meant by an **embedded system**, using the washing machine as an example.

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..... [2]

MARK SCHEMES

Mark Scheme for Question 1 | Source: 9618_w24_ms_12 (Q2.a)

2(a)	1 mark for each bullet point (max 3) e.g. • It is difficult to change / update the firmware by the user • Errors cannot be fixed easily // Troubleshooting / fault-finding / repairing is a specialist task / expensive • Functionality cannot be changed/extended easily // Cannot be easily adapted for another task • Faulty / outdated devices are often thrown away rather than repaired • ... leading to e-waste	3
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Mark Scheme for Question 2 | Source: 9618_s24_ms_11 (Q2.a)

2(a)	1 mark each to max 2 : • The doorbell only performs the specific tasks of motion detection/video recording/doorbell ringing • The motion sensor and digital camera are built into the doorbell • The CPU/memory/storage/software are all dedicated to this task only • Only a dedicated microprocessor is required due to the limited processing requirements	2
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Mark Scheme for Question 3 | Source: 9618_w23_ms_11 (Q9.b)

9(b)	One mark for each reason to max 3 to match the example given • Dedicated to one task applied to example • Does not require much processing power applied to example • Built into a larger system applied to example • Contains firmware that cannot be easily updated applied to example • The system does not have its own operating system • An embedded system must contain a processor, memory and an I/O capability // Dedicated hardware	3
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Mark Scheme for Question 4 | Source: 9618_w23_ms_12 (Q1.c.i)

1(c)(i)	1 mark for each feature (max 2) • Dedicated to a single task // limited number of functions • Built into a larger system // integrated into a larger system • An embedded system must contain a processor, memory and an I/O capability // Dedicated hardware	2
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Mark Scheme for Question 5 | Source: 9618_w23_ms_12 (Q1.c.ii)

1(c)(ii)	1 mark for each bullet point (max 1) - Difficult to change / update the firmware by the user // Difficult to upgrade devices to take advantage of new technology - Cannot be easily adapted for another task - Troubleshooting faults/upgrading is a specialist task - Devices are often thrown away rather than repaired as difficult to upgrade or repair	1
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Mark Scheme for Question 6 | Source: 9618_w22_ms_11 (Q9.a)

9(a)	1 mark for each bullet point (max 2): - the embedded system is built into / integrated into the TV - combination of hardware and software designed for a specific function - The system is not easily changed/updated by the TV owner	2
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Mark Scheme for Question 7 | Source: 9618_w22_ms_13 (Q10.b.ii)

10(b)(ii)	1 mark for each bullet point (max 3): - the embedded system is built into / integrated (into the car alarm) - combination of hardware and software designed for a specific function - must have a processor, memory and input / output - The system is not easily changed/updated by the car owner	3
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Mark Scheme for Question 8 | Source: 9618_s21_ms_11 (Q5.a)

5(a)	1 mark per bullet point to max 2 - Definition: Microprocessor/microcontroller within a larger system // microprocessor/microcontroller that performs one specific task - Example: e.g. Embedded system in washing machine only controls the programs for the washing cycle // it is part of the washing machine but does not perform any other function within it	2
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